

# EXCEL COLUMN CHART

Compare values across categories

01

## WHEN TO USE

- Data has **one category** and **one number** (e.g., Product + Sales).
- Answers: **Which category is highest or lowest?**
- Use it for **rankings** and quick comparisons of about 3-10 categories.

## BUSINESS QUESTION

- Which product generated the highest sales?
- Which product is underperforming?
- How do the products rank against each other?

## DATA EXAMPLE

|   | A              | B            |
|---|----------------|--------------|
| 1 | <b>Product</b> | <b>Sales</b> |
| 2 | Laptop         | 120,000      |
| 3 | Monitor        | 85,000       |
| 4 | Keyboard       | 45,000       |
| 5 | Mouse          | 30,000       |

## DATA ANALYST USE CASE

### Example: Sales Analysis

#### QUESTION

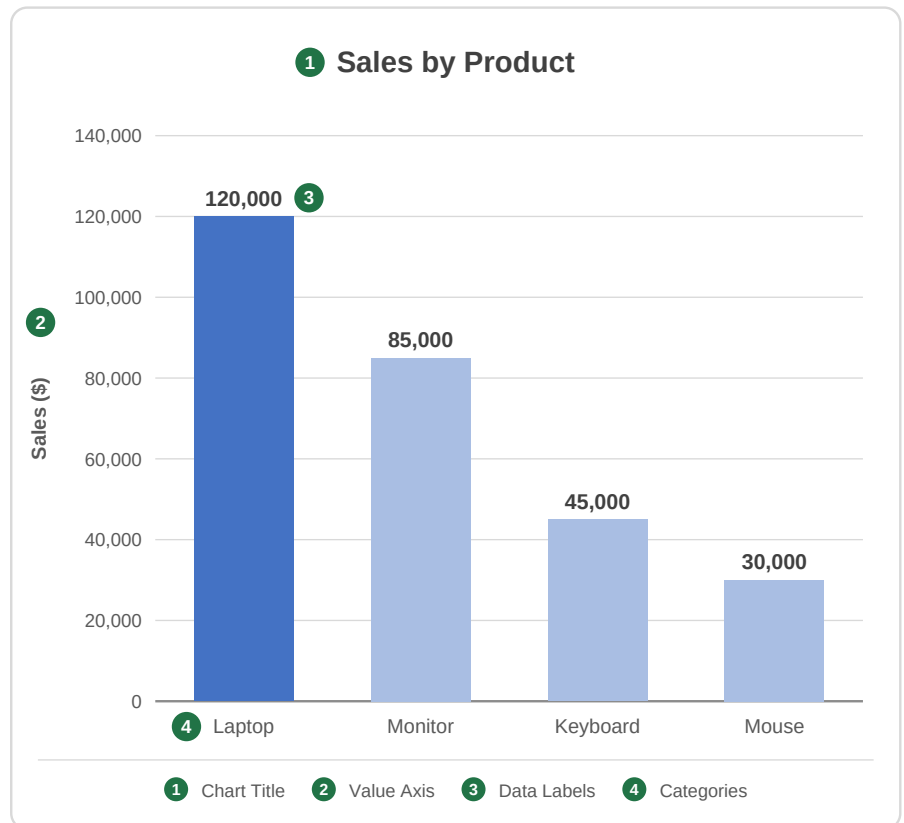
Which product sells the most?

#### CHART

Column Chart

#### INSIGHT

Laptop sales (120,000) are significantly higher than the other products, about 43% of all sales.



## HOW TO CREATE IT IN EXCEL



- Select** the data, including the headers (A1:B5).
- Go to the **Insert** tab and open the **Charts** group.
- Click **Insert Column or Bar Chart** > **2-D Column** > **Clustered Column** (one series = a simple column chart).
- Add a clear **Chart Title** (Chart Design > Add Chart Element > Chart Title).
- Format**: sort data largest to smallest, add Data Labels, use thousand separators.

## DO / DON'T

### DO ✓

- ✓ Use a clear, specific title
- ✓ Sort columns from largest to smallest
- ✓ Keep category labels short and readable
- ✓ Highlight the key column

### DON'T ✗

- ✗ Use a vague title like "Chart 1"
- ✗ Leave categories in random order
- ✗ Overcrowd with 15+ columns
- ✗ Add 3-D effects or gradients



### DATA ANALYST TIP

**Use it when the goal is to compare categories.**

Sort first, label clearly, and highlight the column that carries your message.

# EXCEL BAR CHART

Compare categories, especially long names

02

## WHEN TO USE

- Data has **one category** and **one number**, and the category names are **long**.
- Answers: **Which category is highest or lowest?**
- Use it when names would be **cramped or rotated** on a column chart, or for ranked lists.

## DATA EXAMPLE

|   | A                    | B            |
|---|----------------------|--------------|
| 1 | <b>Department</b>    | <b>Sales</b> |
| 2 | Enterprise Sales     | 480,000      |
| 3 | Consumer Electronics | 360,000      |
| 4 | Online Marketplace   | 290,000      |
| 5 | Corporate Services   | 175,000      |

## DATA ANALYST USE CASE

Example: **Department Performance**

### QUESTION

Which department sells the most?

### CHART

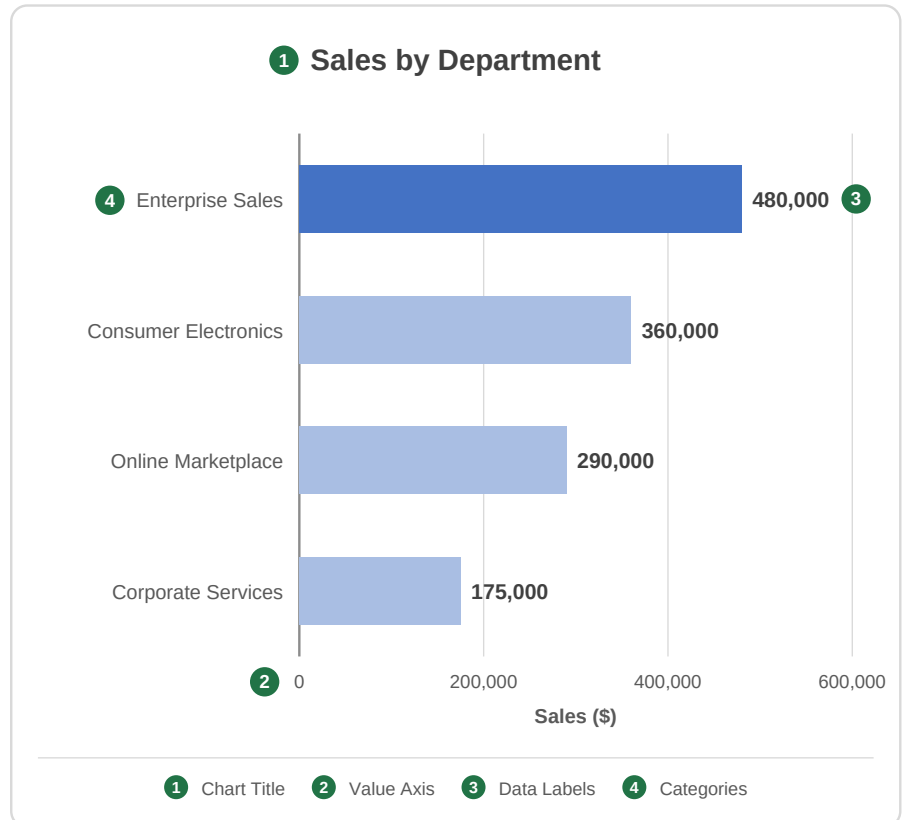
Bar Chart

### INSIGHT

Enterprise Sales leads at 480,000, more than 2.7x the sales of Corporate Services (175,000).

## BUSINESS QUESTION

- Which department generates the most sales?
- Which departments fall behind the others?
- How do departments rank from top to bottom?



## HOW TO CREATE IT IN EXCEL



- Select** the data, including the headers (A1:B5).
- Go to the **Insert** tab and open the **Charts** group.
- Click **Insert Column or Bar Chart** > **2-D Bar** > **Clustered Bar**.
- Add a **Chart Title**. Then **Format Axis** > **Categories in reverse order** so the largest bar is on top.
- Format**: add Data Labels at the bar ends and remove gridlines if labels show the values.

## DO / DON'T

### DO ✓

- ✓ Use bars when category names are long
- ✓ Sort so the largest bar is on top
- ✓ Start the value axis at zero
- ✓ Label values at the end of each bar

### DON'T ✗

- ✗ Rotate long labels on a column chart
- ✗ Leave bars unsorted
- ✗ Truncate the axis to exaggerate gaps
- ✗ Use many colours for one series



### DATA ANALYST TIP

**When category names are long, switch from Column to Bar.**

*Horizontal bars read like a ranked list and keep every label readable.*

# EXCEL LINE CHART

Show trends over time

# 03

## WHEN TO USE

- Data has a **time field** (day, month, year) and **one number**.
- Answers: **Is it going up, down, or staying flat?**
- Use it for **trends, growth and seasonality** over time.

## BUSINESS QUESTION

- How has monthly revenue changed this year?
- Which month had the highest revenue?
- Is revenue growing or declining?

## DATA EXAMPLE

|   | A     | B       |
|---|-------|---------|
| 1 | Month | Revenue |
| 2 | Jan   | 42,000  |
| 3 | Feb   | 45,500  |
| 4 | Mar   | 51,000  |
| 5 | Apr   | 48,500  |
| 6 | May   | 58,000  |
| 7 | Jun   | 66,500  |

## DATA ANALYST USE CASE

### Example: Monthly Revenue

#### QUESTION

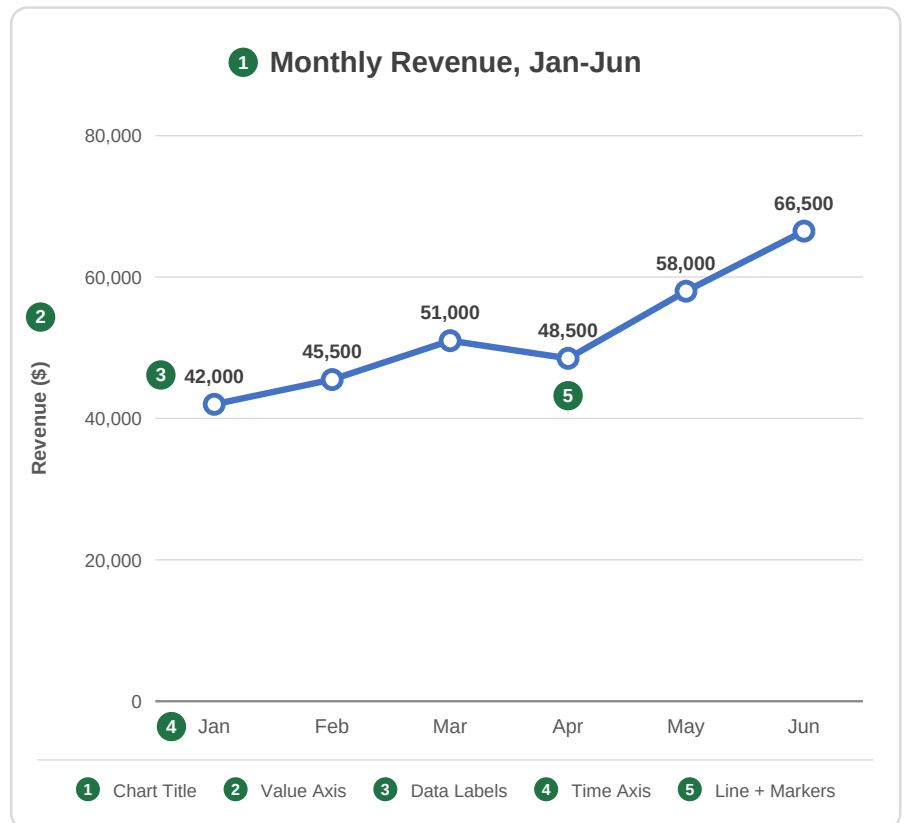
Is revenue growing?

#### CHART

Line Chart

#### INSIGHT

Revenue rose from 42,000 in January to 66,500 in June (+58%), with a small dip in April.



## HOW TO CREATE IT IN EXCEL



- 1 **Select** the data, including the headers (A1:B7).
- 2 Go to the **Insert** tab and open the **Charts** group.
- 3 Click **Insert Line or Area Chart > 2-D Line > Line with Markers**.
- 4 Add a clear **Chart Title** that names the measure and time period.
- 5 **Format**: keep time in order, add labels for key points, and start the axis at a sensible value.

## DO / DON'T

### DO ✓

- ✓ Put time on the horizontal axis
- ✓ Keep dates in chronological order
- ✓ Use one clear line per measure
- ✓ Mark the peaks and key points

### DON'T ✗

- ✗ Use it for unordered categories
- ✗ Plot more than 4-5 lines
- ✗ Skip months or mix time intervals
- ✗ Use 3-D or heavy effects



### DATA ANALYST TIP

**Use it when time and trend matter.**

*If the horizontal axis is not time, a column or bar chart is usually the better choice.*

# EXCEL PIE CHART

Show part-to-whole relationships

04

## WHEN TO USE

- Data has **2-5 categories** that add up to a **100% total**.
- Answers: **What share does each part contribute?**
- Use it for a **simple share message** when exact comparison is not needed.

## BUSINESS QUESTION

- What share of revenue does each product bring?
- Which product contributes the most to revenue?
- Do two products make up most of the total?

## DATA EXAMPLE

|   | A              | B              |
|---|----------------|----------------|
| 1 | <b>Product</b> | <b>Revenue</b> |
| 2 | Laptop         | 120,000        |
| 3 | Monitor        | 85,000         |
| 4 | Keyboard       | 45,000         |
| 5 | Mouse          | 30,000         |

## DATA ANALYST USE CASE

Example: **Revenue Share**

### QUESTION

How is revenue split across products?

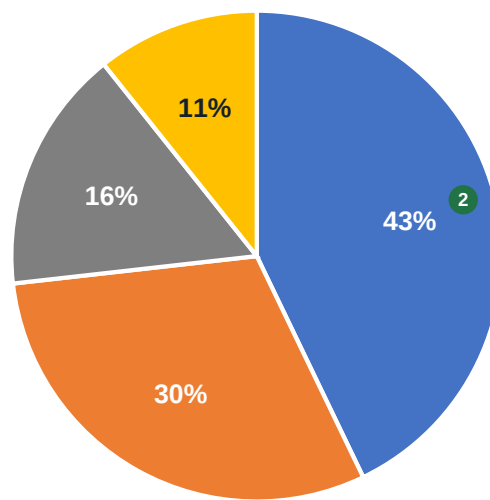
### CHART

Pie Chart

### INSIGHT

Laptops and Monitors together bring in about 73% of total revenue.

## 1 Revenue Share by Product



3 Laptop Monitor Keyboard Mouse

1 Chart Title 2 Data Labels (%) 3 Legend = Categories

## HOW TO CREATE IT IN EXCEL

Select Data

Insert

Pie Chart

Add Title

Format

Analyze

- Select the data, including the headers (A1:B5).
- Go to the **Insert** tab and open the **Charts** group.
- Click **Insert Pie or Doughnut Chart > 2-D Pie > Pie**.
- Add a clear **Chart Title** that states what the 100% represents.
- Format:** add Data Labels showing **Percentage**, order slices largest first, keep the legend.

## DO / DON'T

### DO ✓

- ✓ Limit the chart to 5 slices or fewer
- ✓ Show percentages as data labels
- ✓ Order slices from largest to smallest
- ✓ Make sure the parts add up to 100%

### DON'T ✗

- ✗ Use it for many categories
- ✗ Compare slices of similar size
- ✗ Use it to show trends over time
- ✗ Explode slices or use 3-D

### DATA ANALYST TIP



**Use it only when showing a simple part-to-whole relationship.**

*If the slices look alike in size, switch to a bar chart so differences are visible.*

# EXCEL DOUGHNUT CHART

Part-to-whole in a dashboard-friendly style

# 05

## WHEN TO USE

- Data has **2-5 categories** that add up to a **whole**.
- Answers: **How is the total divided?**
- Use it on **dashboards**: the centre hole can show a key total or KPI.

## BUSINESS QUESTION

- What share of customers is in each segment?
- Which segment is our biggest customer group?
- How big is Enterprise compared with the rest?

## DATA EXAMPLE

|   | A              | B                |
|---|----------------|------------------|
| 1 | <b>Segment</b> | <b>Customers</b> |
| 2 | Consumer       | 5,200            |
| 3 | Small Business | 2,600            |
| 4 | Enterprise     | 1,200            |
| 5 | Education      | 1,000            |

## DATA ANALYST USE CASE

Example: **Customer Segmentation**

### QUESTION

Who are our customers?

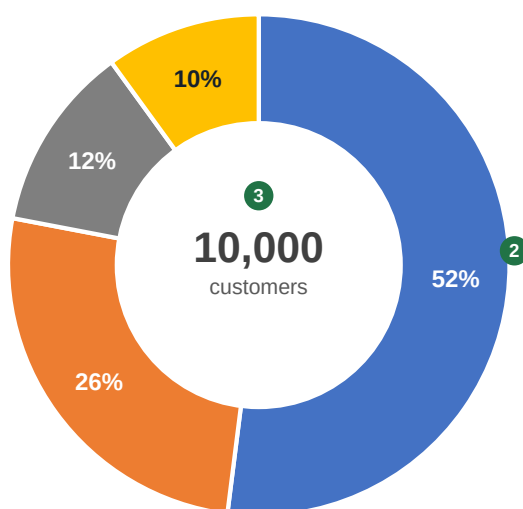
### CHART

Doughnut Chart

### INSIGHT

Consumers make up 52% of customers, while Enterprise is only 12%.

## 1 Customer Segment Share



4 Consumer 26% Small Business 12% Enterprise 10% Education

1 Chart Title 2 Data Labels (%) 3 Centre Total 4 Legend

## HOW TO CREATE IT IN EXCEL



- Select the data, including the headers (A1:B5).
- Go to the **Insert** tab and open the **Charts** group.
- Click **Insert Pie or Doughnut Chart > Doughnut**.
- Add a **Chart Title**, then **Format Data Series > Doughnut Hole Size** (about 55%).
- Format**: add percentage labels and place the total in a centre text box.

## DO / DON'T

### DO ✓

- ✓ Keep to 5 segments or fewer
- ✓ Show percentages on the ring
- ✓ Use the centre for the total or KPI
- ✓ Use consistent, distinct colours

### DON'T ✗

- ✗ Use many thin slices
- ✗ Rely on the ring to compare sizes
- ✗ Use it for trends over time
- ✗ Use clashing or neon colours



### DATA ANALYST TIP

**Use a doughnut when the total matters, and put it in the centre.**

*Great for dashboards. When exact comparison matters, use a bar chart instead.*

# EXCEL STACKED COLUMN CHART

06

Show totals and what they are made of

## WHEN TO USE

- Data has **categories** split into **components** (e.g., Month x Product Category).
- Answers: **How big is the total, and what makes it up?**
- Use it to compare **totals over time** while keeping the composition visible.

## DATA EXAMPLE

|   | A     | B       | C           | D        |
|---|-------|---------|-------------|----------|
| 1 | Month | Laptops | Accessories | Software |
| 2 | Jan   | 60      | 25          | 15       |
| 3 | Feb   | 65      | 30          | 15       |
| 4 | Mar   | 80      | 35          | 20       |
| 5 | Apr   | 75      | 40          | 25       |

## DATA ANALYST USE CASE

Example: **Sales Mix**

### QUESTION

What drives monthly sales growth?

### CHART

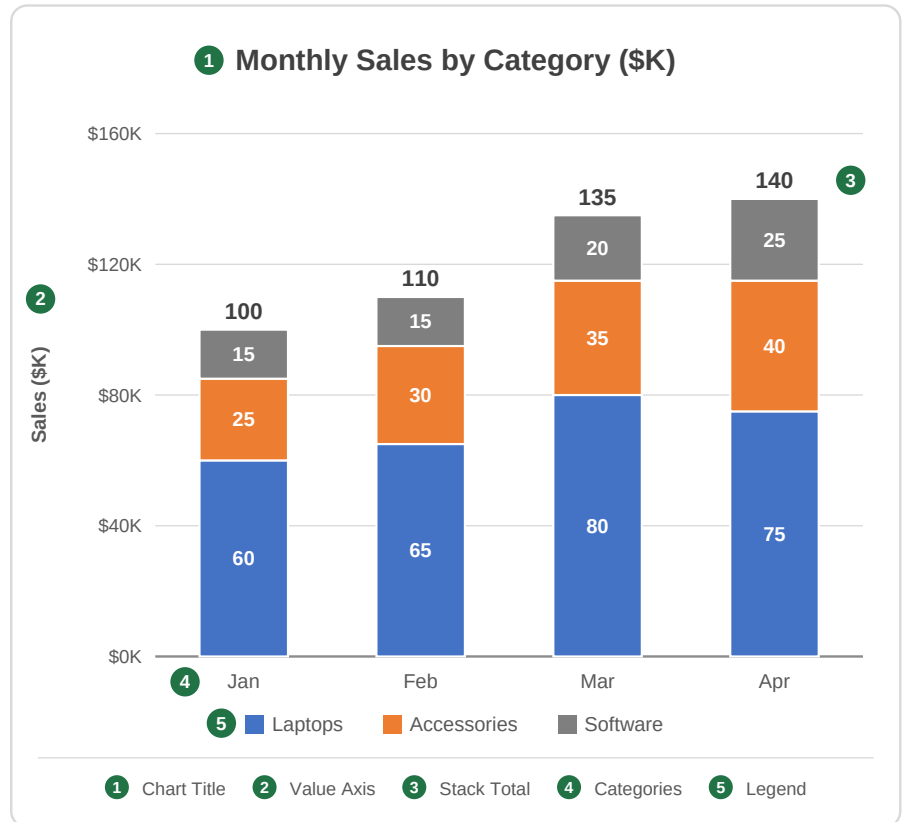
Stacked Column Chart

### INSIGHT

Sales grew 40% (100K to 140K), led equally by Laptops and Accessories (+15K each).

## BUSINESS QUESTION

- How did total monthly sales change?
- Which category drives each month's sales?
- Is Software growing over time?



## HOW TO CREATE IT IN EXCEL



- Select** all the data, including the headers (A1:D5).
- Go to the **Insert** tab and open the **Charts** group.
- Click **Insert Column or Bar Chart** > **2-D Column** > **Stacked Column**.
- Add a clear **Chart Title** and move the legend to the bottom.
- Format**: add Data Labels, keep colours consistent, and put the key series at the bottom.

## DO / DON'T

### DO ✓

- Limit the stack to 3-4 categories
- Use one colour per category
- Put the key series at the bottom
- Show the total above each stack

### DON'T ✗

- Stack 8+ categories
- Compare middle segments precisely
- Use similar-looking colours
- Use it when only totals matter



### DATA ANALYST TIP

**Use it to show total + composition, and keep the key series at the bottom.**

*Only the bottom segment shares a common baseline, so order your series with care.*

# EXCEL STACKED BAR CHART

Show composition across categories with long names

07

## WHEN TO USE

- Data has **long category names** split into **components**.
- Answers: **How is each group made up?**
- Use it for **headcount**, **survey results** or any breakdown by group.

## BUSINESS QUESTION

- How many employees does each department have?
- What is the gender split in each department?
- Which department is the largest?

## DATA EXAMPLE

|   | A                      | B    | C      |
|---|------------------------|------|--------|
| 1 | Department             | Male | Female |
| 2 | Information Technology | 58   | 32     |
| 3 | Customer Support       | 30   | 45     |
| 4 | Finance                | 20   | 22     |
| 5 | Human Resources        | 12   | 28     |

## DATA ANALYST USE CASE

Example: **Workforce Analysis**

### QUESTION

How is each department composed?

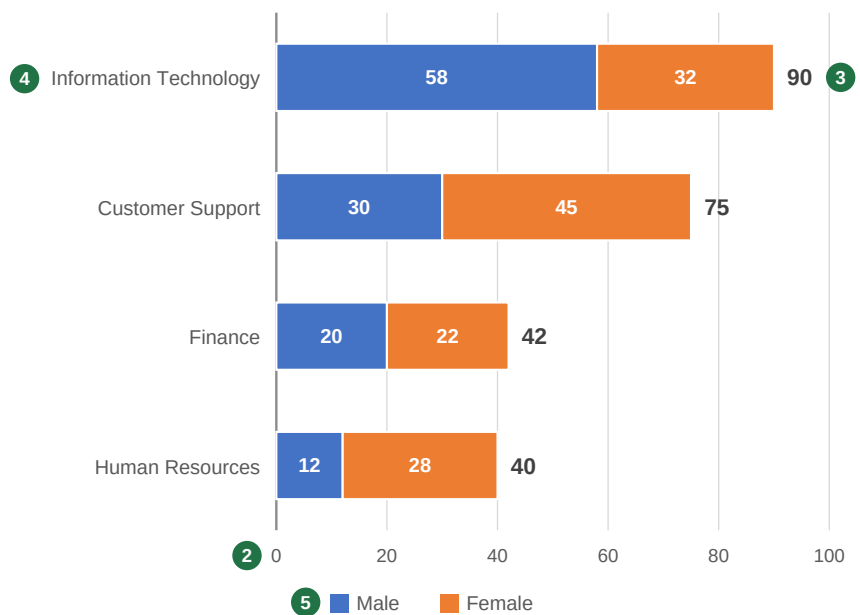
### CHART

Stacked Bar Chart

### INSIGHT

IT is the largest department (90 staff, 64% male); HR is mostly female (70%).

## 1 Employee Count by Department & Gender



1 Chart Title 2 Value Axis 3 Bar Total 4 Categories 5 Legend

## HOW TO CREATE IT IN EXCEL



- Select** all the data, including the headers (A1:C5).
- Go to the **Insert** tab and open the **Charts** group.
- Click **Insert Column or Bar Chart** > **2-D Bar** > **Stacked Bar**.
- Add a **Chart Title**, then reverse the category order so the first department is on top.
- Format**: add Data Labels inside the bars and keep the legend at the bottom.

## DO / DON'T

### DO ✓

- ✓ Use bars for long category names
- ✓ Keep to 2-4 segments
- ✓ Add data labels inside the bars
- ✓ Show the total at the bar end

### DON'T ✗

- ✗ Use too many segments
- ✗ Compare non-baseline segments exactly
- ✗ Leave the bars unsorted
- ✗ Use it for a single series

### DATA ANALYST TIP



**Use stacked bars for composition across categories with long names.**

*If you need to compare segments exactly, use a clustered bar chart instead.*

# EXCEL CLUSTERED COLUMN CHART

08

Compare multiple series side by side

## WHEN TO USE

- Data has **categories** and **2-3 measures** to compare (e.g., Actual vs Target).
- Answers: **Which series is higher in each category?**
- Use it for **performance vs goal**, this year vs last year, or group comparisons.

## BUSINESS QUESTION

- Which regions met their sales target?
- How far above or below target is each region?
- Which region missed its target the most?

## DATA EXAMPLE

|   | A             | B             | C             |
|---|---------------|---------------|---------------|
| 1 | <b>Region</b> | <b>Target</b> | <b>Actual</b> |
| 2 | North         | 100           | 112           |
| 3 | South         | 90            | 78            |
| 4 | East          | 80            | 85            |
| 5 | West          | 70            | 64            |

## DATA ANALYST USE CASE

Example: **Target Tracking**

### QUESTION

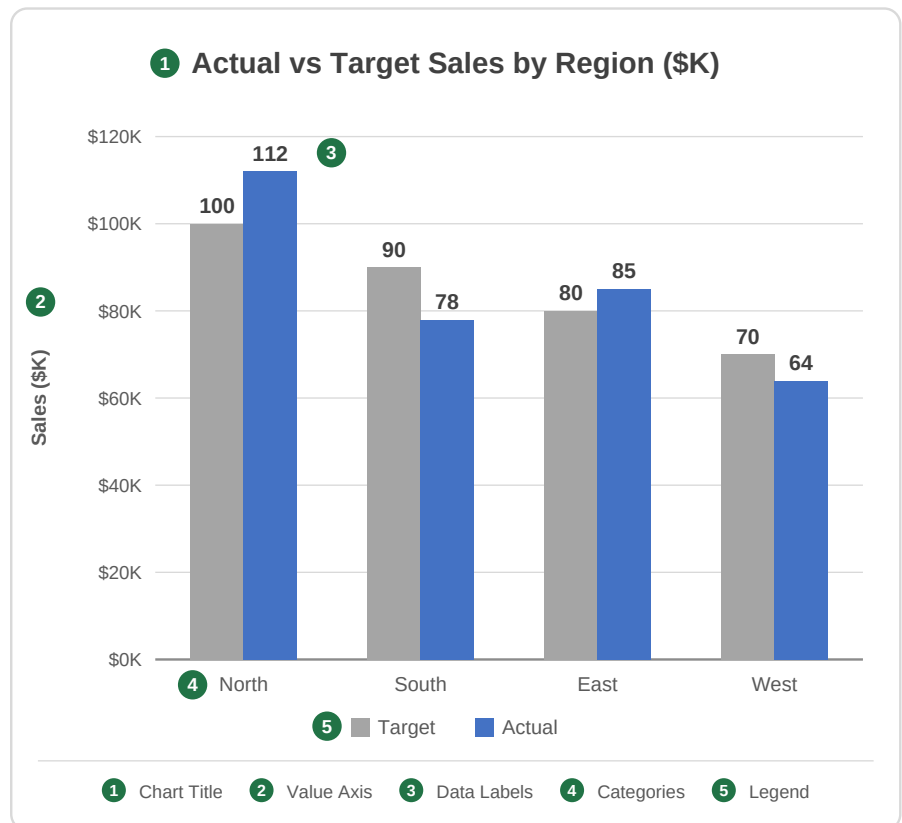
Did each region hit its target?

### CHART

Clustered Column Chart

### INSIGHT

North (+12%) and East (+6%) beat target; South (-13%) and West (-9%) fell short.



## HOW TO CREATE IT IN EXCEL



- 1** Select all the data, including the headers (A1:C5).
- 2** Go to the **Insert** tab and open the **Charts** group.
- 3** Click **Insert Column or Bar Chart** > **2-D Column** > **Clustered Column**.
- 4** Add a clear **Chart Title** and keep the legend so readers know Target from Actual.
- 5** **Format**: use grey for Target, a strong colour for Actual, and add Data Labels.

## DO / DON'T

### DO ✓

- ✓ Limit to 2-3 series per category
- ✓ Use grey for Target, colour for Actual
- ✓ Keep the series order consistent
- ✓ Add a legend or direct labels

### DON'T ✗

- ✗ Add too many series
- ✗ Use similar-looking colours
- ✗ Cram in more than 6 categories
- ✗ Mix up which series is which

### DATA ANALYST TIP



**Use it when comparing two or three measures within each category.**

*More than three series? Try a line chart, or split it into separate charts.*

# EXCEL COMBO CHART

Compare two different measures in one chart

09

## WHEN TO USE

- Data has **two measures with different units** (e.g., \$ and %).
- Answers: **How do volume and rate move together?**
- Use **Column + Line**, and add a **secondary axis** when the scales differ.

## BUSINESS QUESTION

- Did higher sales also mean higher profit margin?
- In which month was the margin highest?
- Are we growing sales at the cost of margin?

## DATA EXAMPLE

|   | A     | B           | C        |
|---|-------|-------------|----------|
| 1 | Month | Sales (\$K) | Margin % |
| 2 | Jan   | 80          | 12%      |
| 3 | Feb   | 95          | 14%      |
| 4 | Mar   | 110         | 13%      |
| 5 | Apr   | 100         | 11%      |
| 6 | May   | 130         | 16%      |
| 7 | Jun   | 150         | 18%      |

## DATA ANALYST USE CASE

Example: **Sales & Profitability**

### QUESTION

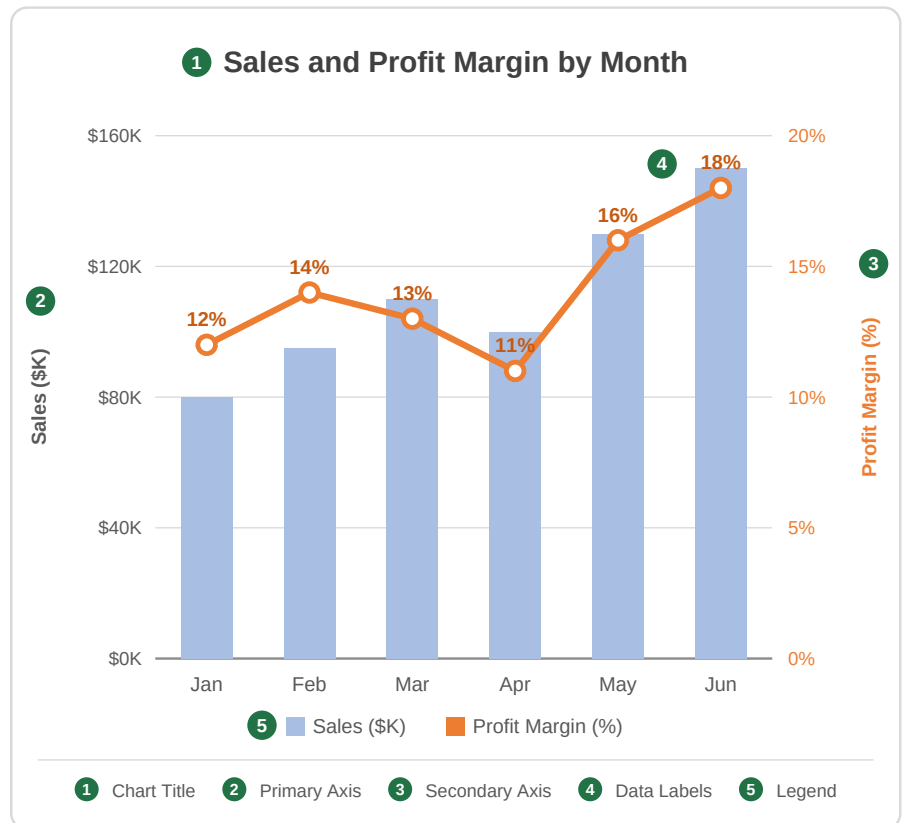
Is our growth profitable?

### CHART

Combo Chart (Column + Line)

### INSIGHT

Sales nearly doubled (80K to 150K) while margin rose from 12% to 18%: growth is profitable.



## HOW TO CREATE IT IN EXCEL



- Select** all the data, including the headers (A1:C7).
- Go to **Insert > Charts** and click **Insert Combo Chart > Create Custom Combo Chart**.
- Set Sales to **Clustered Column**, Margin to **Line**, and tick **Secondary Axis** for Margin.
- Add a **Chart Title** and an axis title for each axis (Sales \$K, Margin %).
- Format**: label only the line, and colour-match each axis to its series.

## DO / DON'T

### DO ✓

- ✓ Use a secondary axis for different units
- ✓ Label both axes clearly
- ✓ Match axis colour to its series
- ✓ Columns for volume, line for rate

### DON'T ✗

- ✗ Combine unrelated measures
- ✗ Leave out the axis titles
- ✗ Use misleading axis scales
- ✗ Add a third measure



### DATA ANALYST TIP

**Use it when two measures with different units tell one story.**

*Always label both axes. A secondary axis without titles confuses readers.*

# EXCEL SCATTER CHART

Explore relationships between two numbers

# 10

## WHEN TO USE

- Data has **two numeric variables** measured for each record.
- Answers: **Do the two variables move together?**
- Use it to spot **correlation, patterns and outliers** before deeper analysis.

## BUSINESS QUESTION

- Does higher advertising spend lead to higher sales?
- Is the relationship positive, negative or none?
- Which points are outliers worth investigating?

## DATA EXAMPLE

|   | A        | B     |
|---|----------|-------|
| 1 | Ad Spend | Sales |
| 2 | 5        | 40    |
| 3 | 8        | 52    |
| 4 | 10       | 58    |
| 5 | 14       | 70    |
| 6 | 17       | 81    |
| 7 | 20       | 88    |
| 8 | 24       | 101   |
| 9 | 28       | 64    |

## DATA ANALYST USE CASE

Example: **Marketing Analysis**

### QUESTION

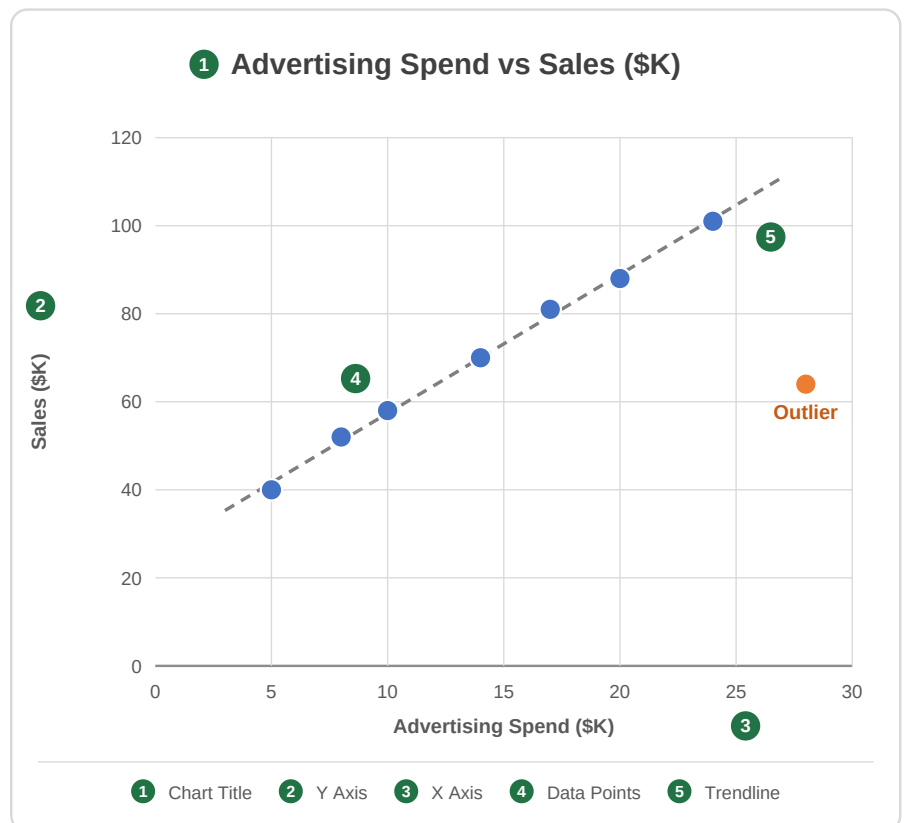
Does advertising drive sales?

### CHART

Scatter Chart

### INSIGHT

Sales rise steadily with ad spend, except one outlier at 28K spend (64K sales) worth investigating.



## HOW TO CREATE IT IN EXCEL



- Select** the two numeric columns, including the headers (A1:B9).
- Go to the **Insert** tab and open the **Charts** group.
- Click **Insert Scatter (X, Y) or Bubble Chart > Scatter** (markers only).
- Add a **Chart Title** and **axis titles** for both variables, with units.
- Format**: add a **Trendline** (Chart Elements > Trendline) and investigate outliers.

## DO / DON'T

### DO ✓

- ✓ Put the driver (X) on the horizontal axis
- ✓ Label both axes with units
- ✓ Add a trendline
- ✓ Investigate outliers

### DON'T ✗

- ✗ Use it for categories
- ✗ Assume correlation means causation
- ✗ Connect the points with lines
- ✗ Ignore outliers



### DATA ANALYST TIP

**Use it to explore relationships between two numerical variables.**

*Correlation is not causation. Use scatter charts to find questions, not final answers.*